Introduction to Machine Learning

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- Why do we need machine learning?
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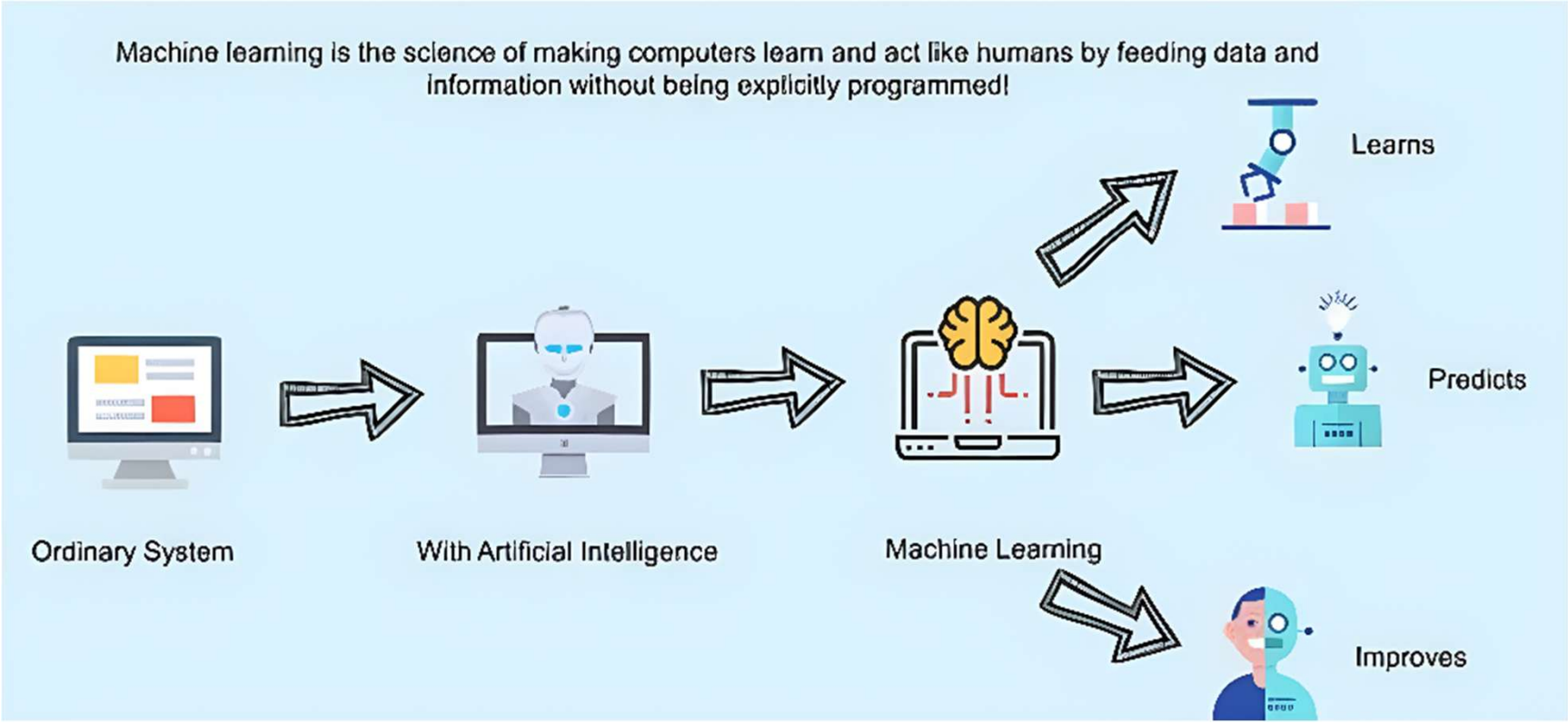
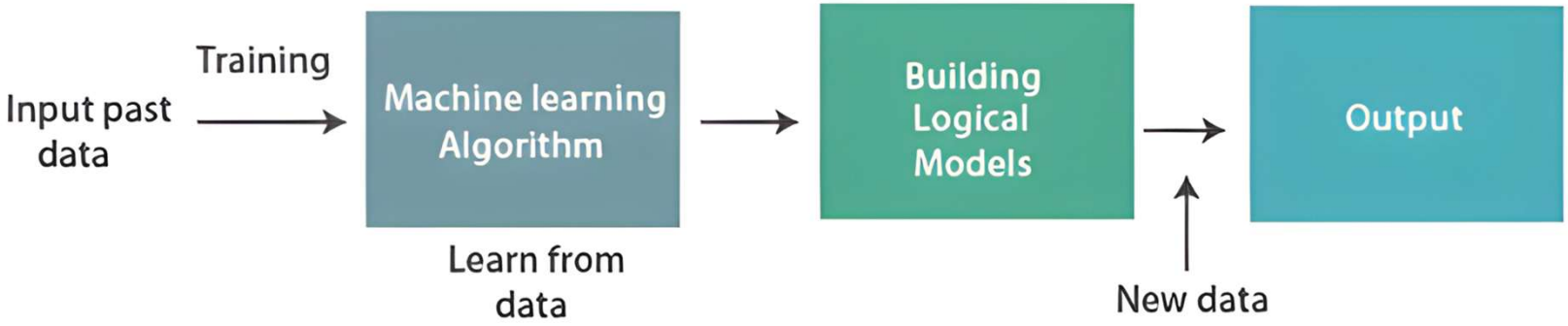
Introduction to Machine Learning



How does it all Work



- The accuracy and number of datapoints affect the accuracy!

Need For Machine Learning

Following are some key points which show the importance of Machine Learning:

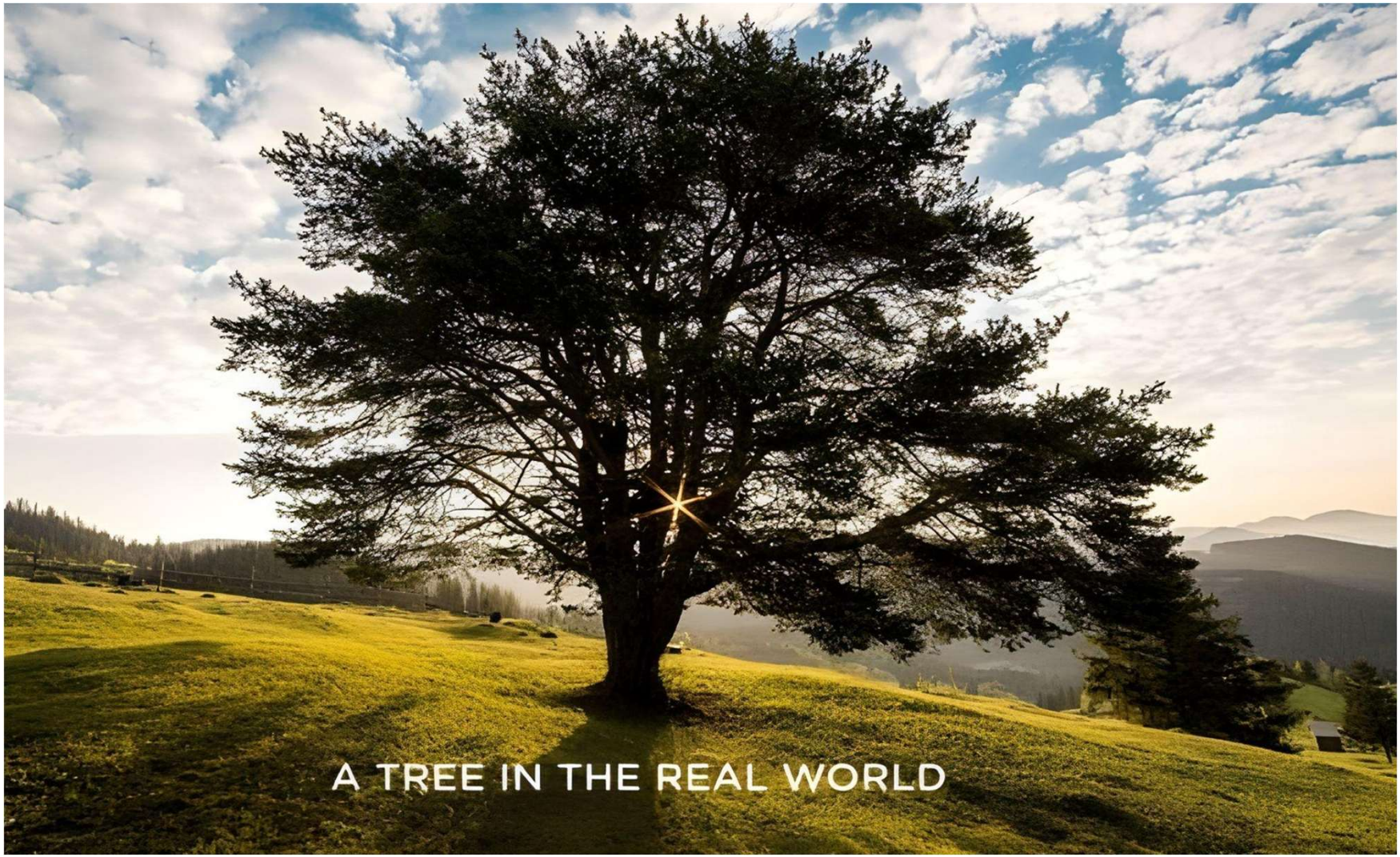
- Rapid increment in the production of data
- Solving complex problems, which are difficult for a human
- Finding **hidden patterns** and **extracting useful information** from data
- To deal with time constraints and the overall cost optimization of **experimental analysis**
- High Throughput of analytical techniques
- Predict outcomes
- Decision making in various sector including **scientific research**

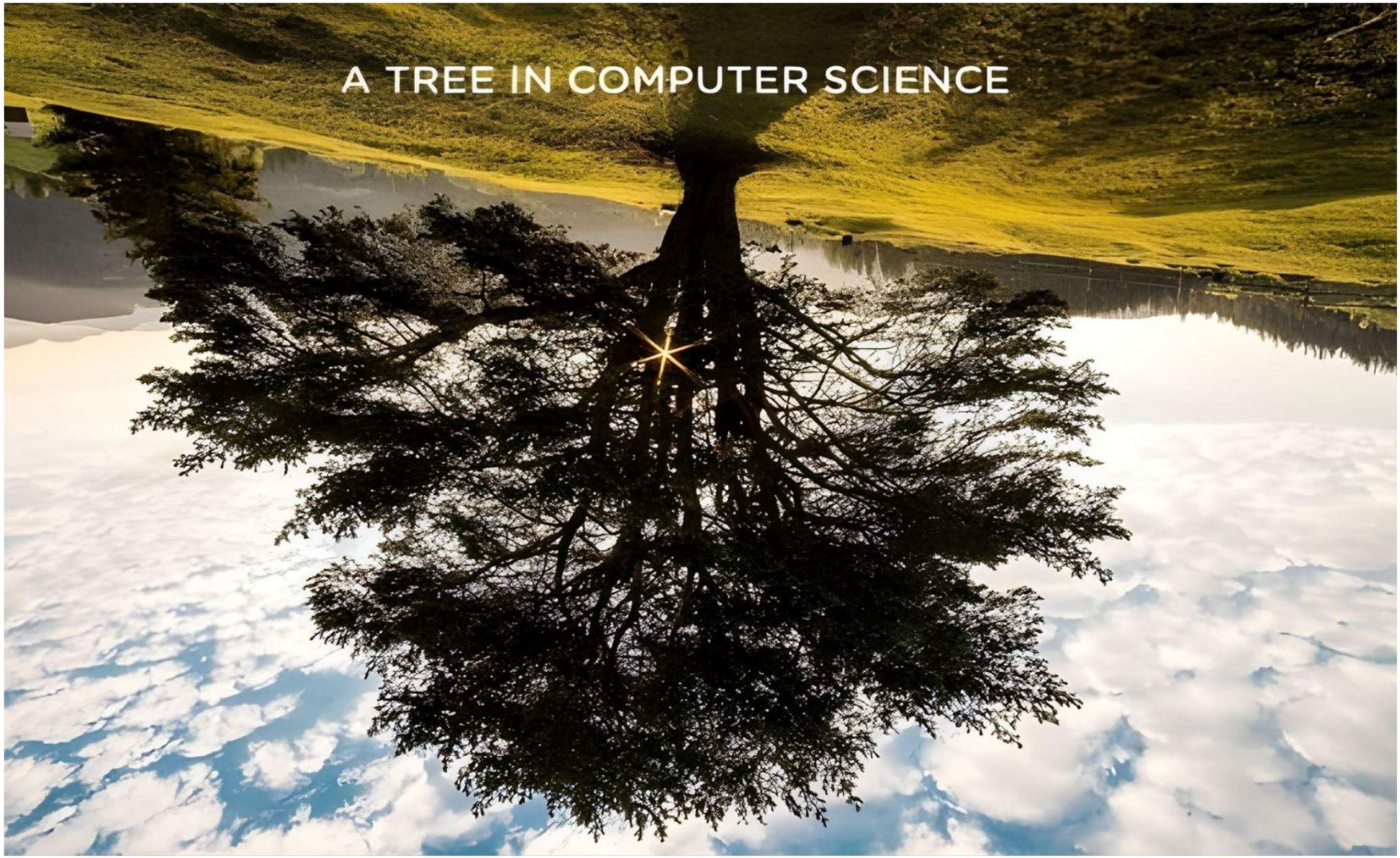
Decision Tree and Random Forest

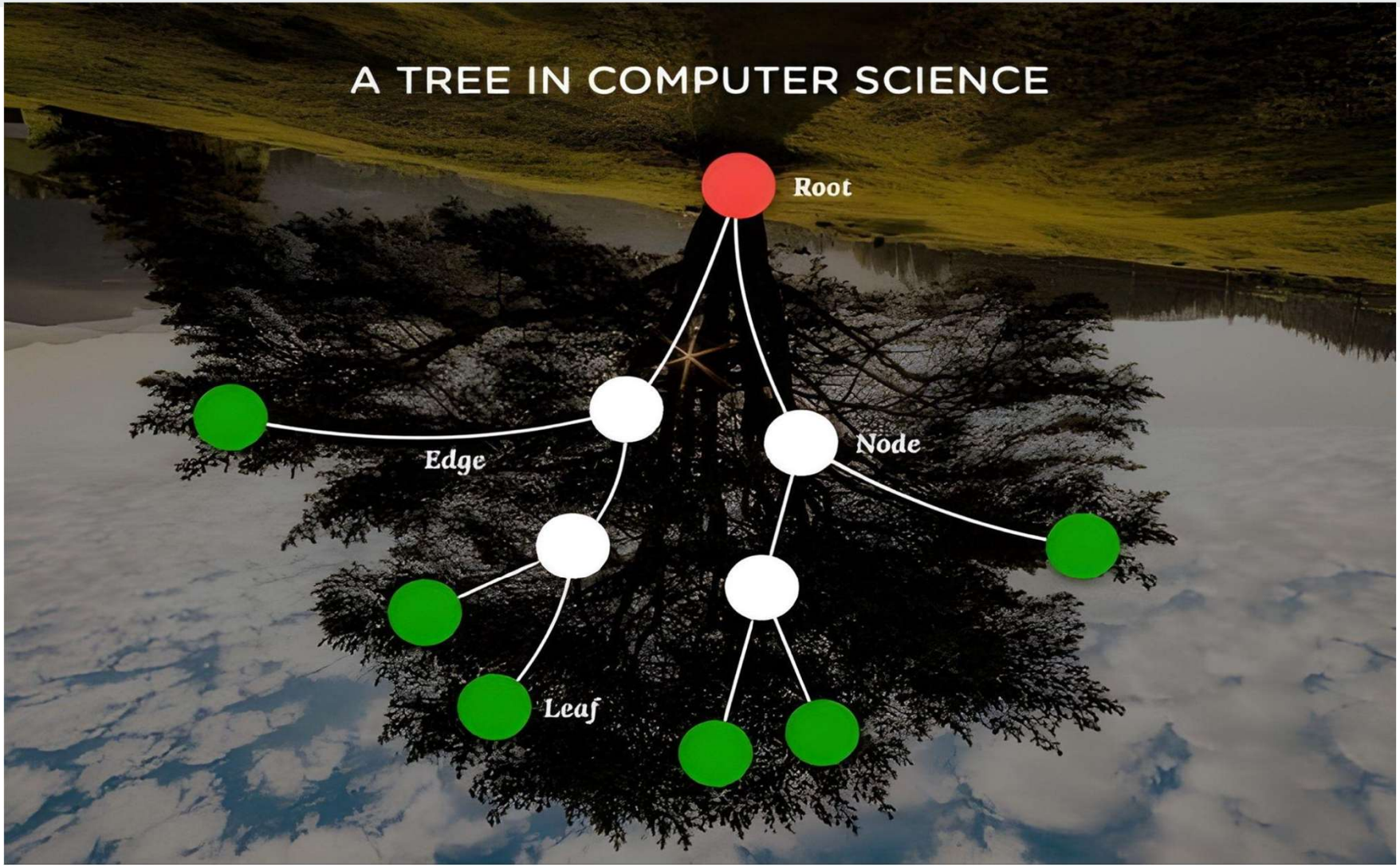
There are many different ways in which machine learning models make decisions.

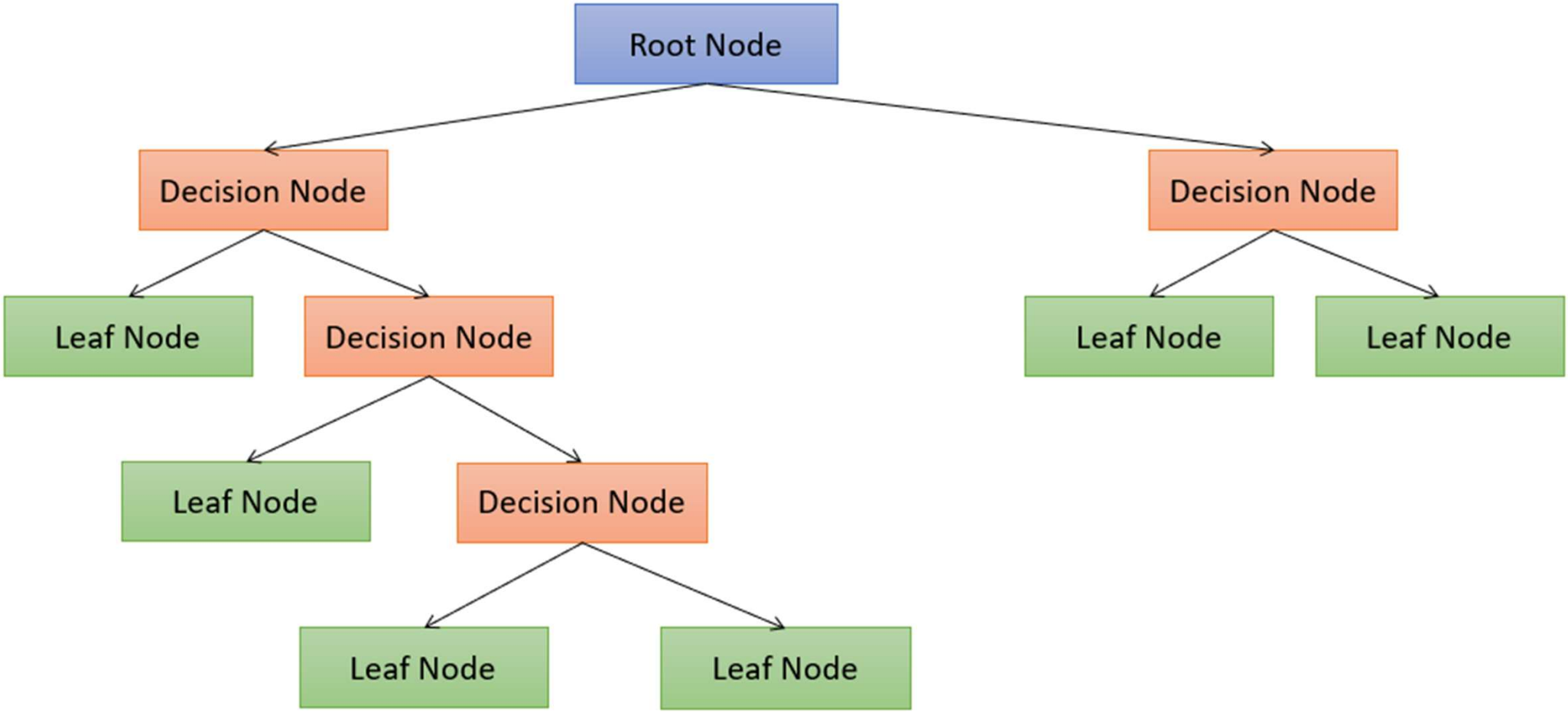
Decision Trees and Random Forests are two of the most common decision-making processes used in Machine Learning.

Decision Tree



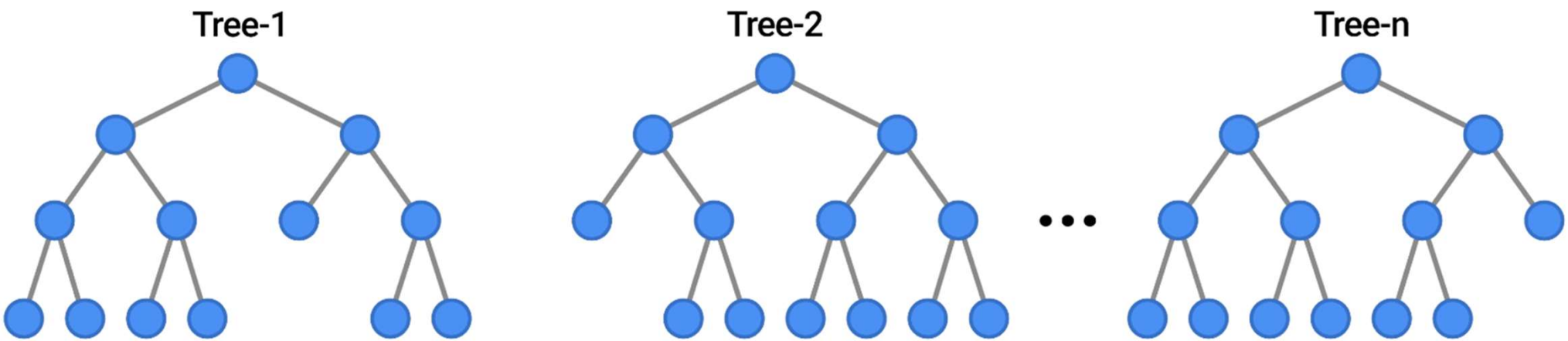

A TREE IN COMPUTER SCIENCE


A TREE IN COMPUTER SCIENCE
Root
Edge
Node
Leaf

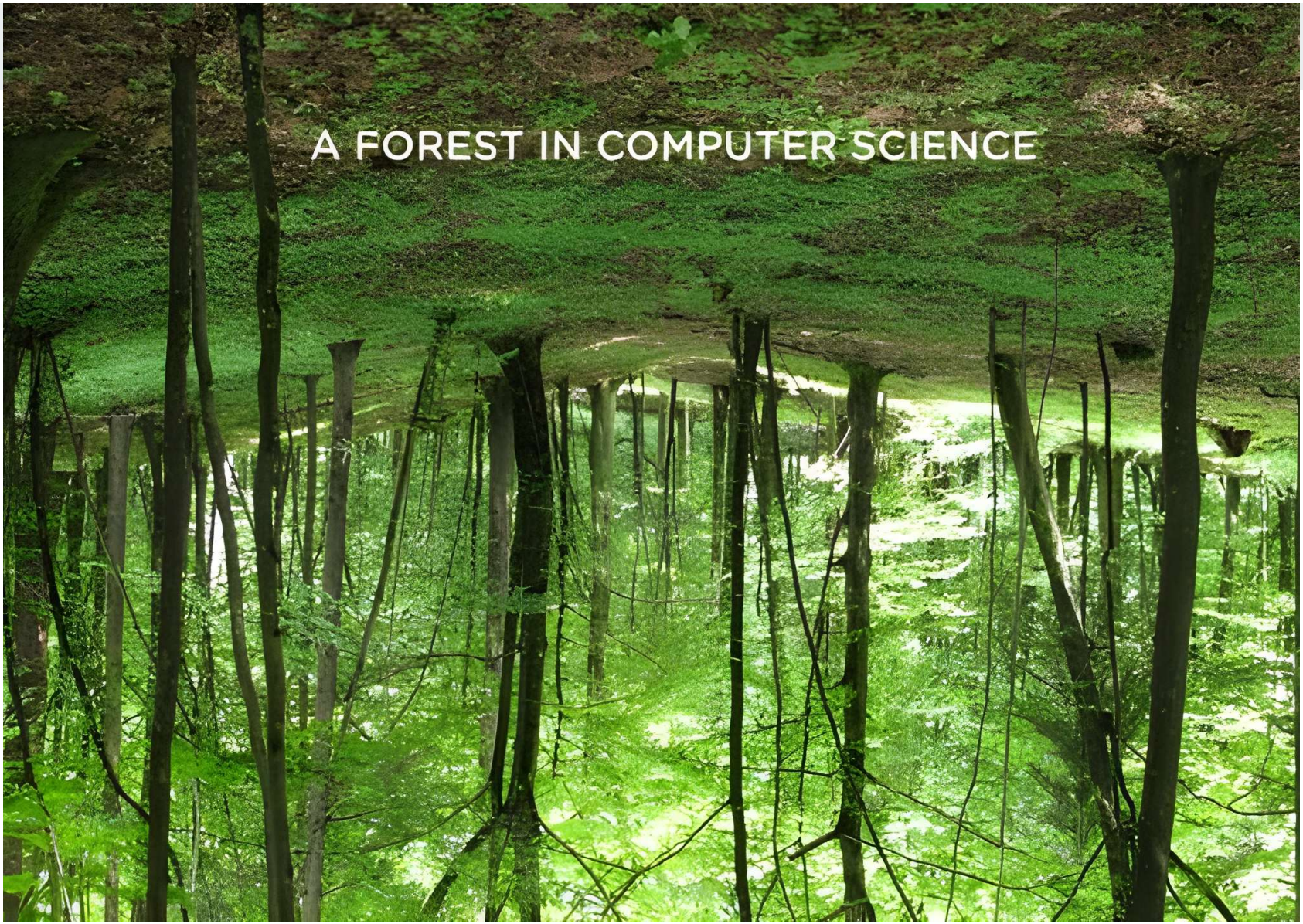

Root Node
Decision Node
Decision Node
Leaf Node
Decision Node
Leaf Node
Leaf Node
Leaf Node
Decision Node
Leaf Node
Leaf Node

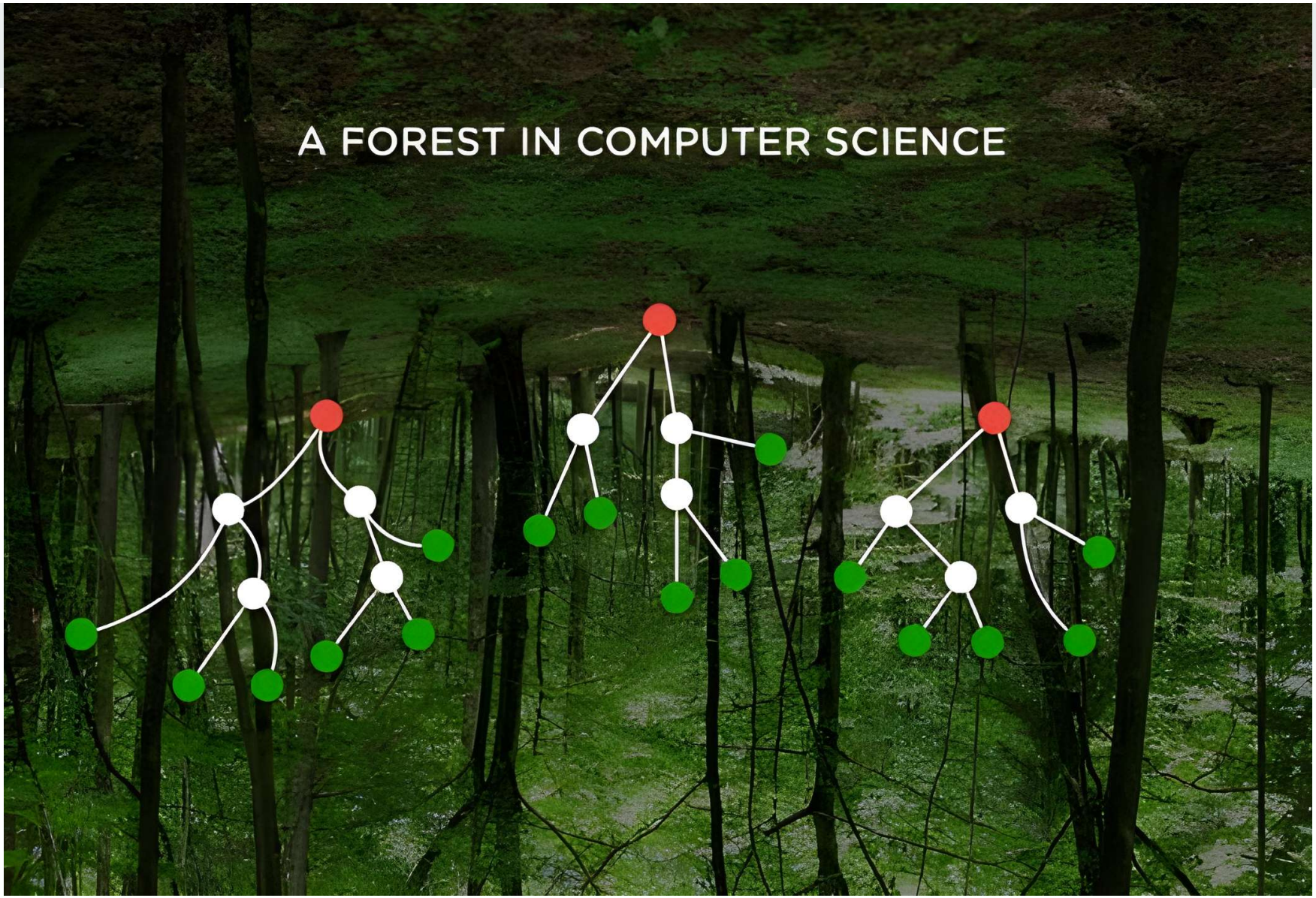
Random Forest

The Random Forest Algorithm combines the output of multiple (randomly created) decision trees to generate the final output.

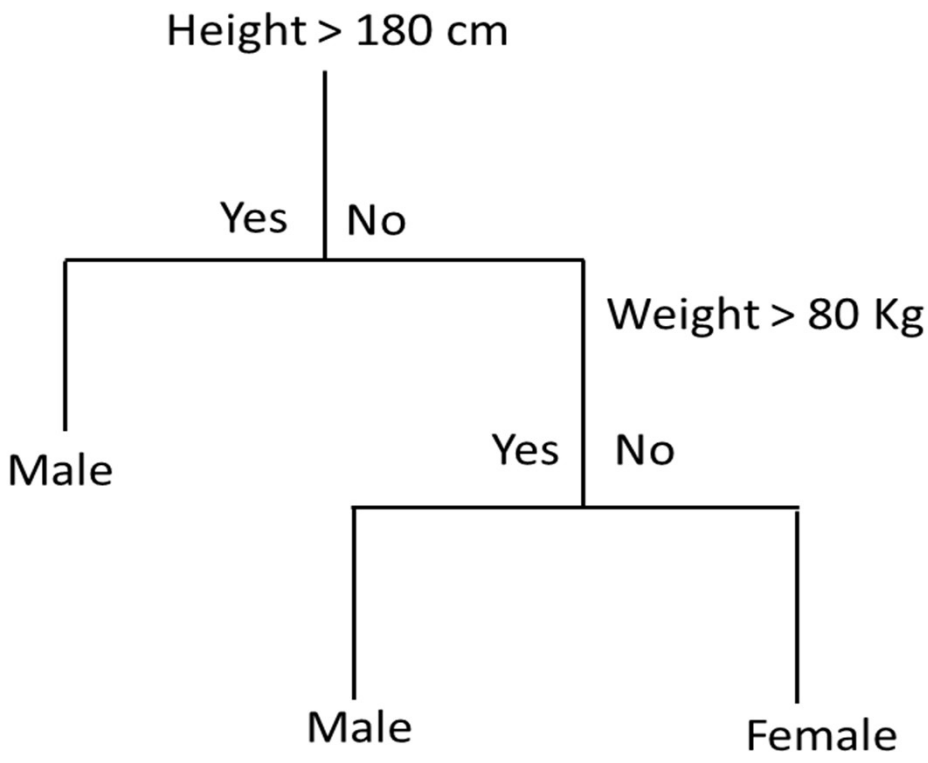


A FOREST IN THE REAL WORLD


A FOREST IN COMPUTER SCIENCE


A FOREST IN COMPUTER SCIENCE

Example (DT)

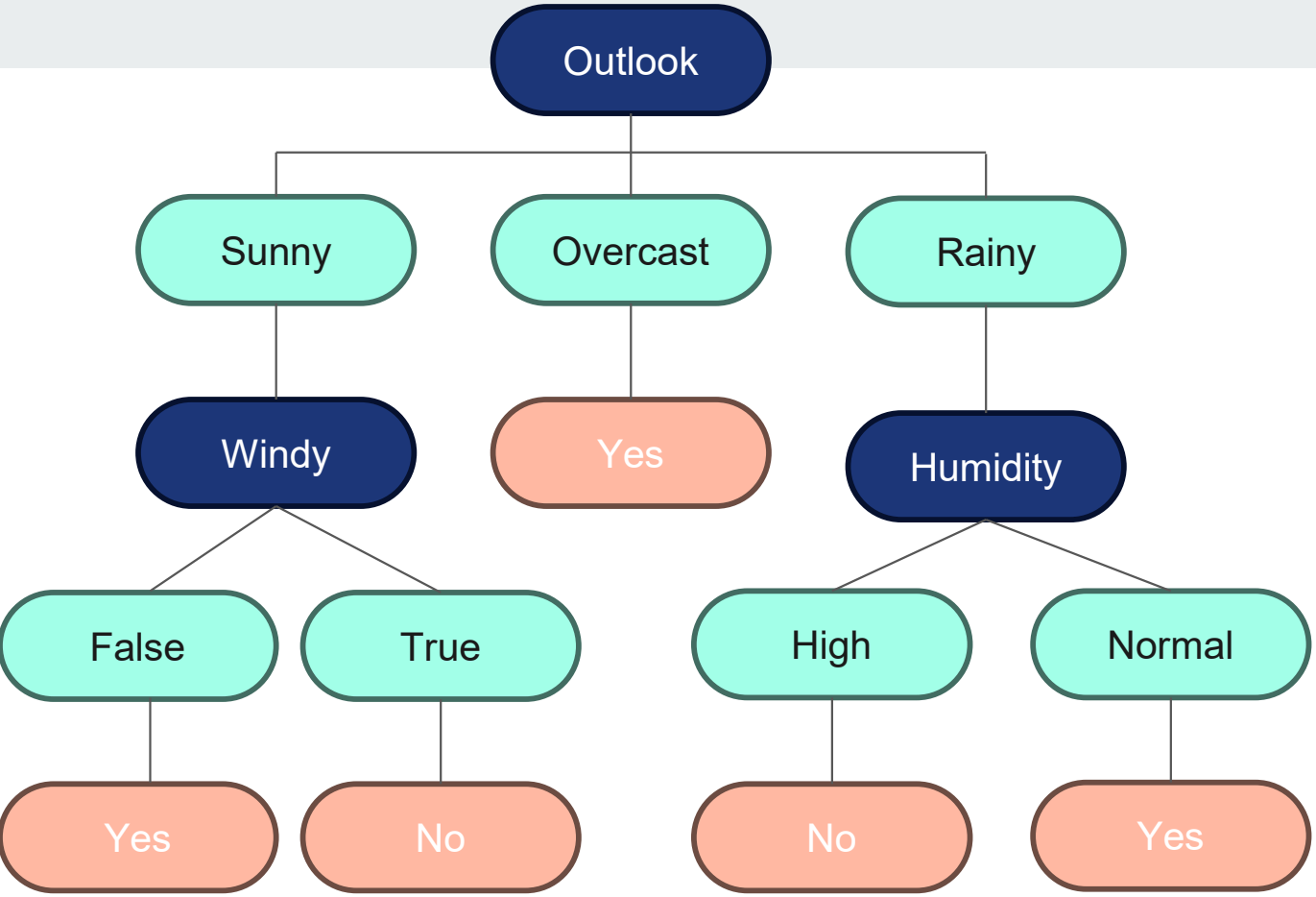


DT

Building a Decision Tree

Let us start with an example:

Outlook	Temp	Humidity	Windy	Play?
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cook	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No



Decision Tree

Outlook	Temp	Humidity	Windy	Play?
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
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Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

Outlook	Temp	Humidity	Windy	Play?
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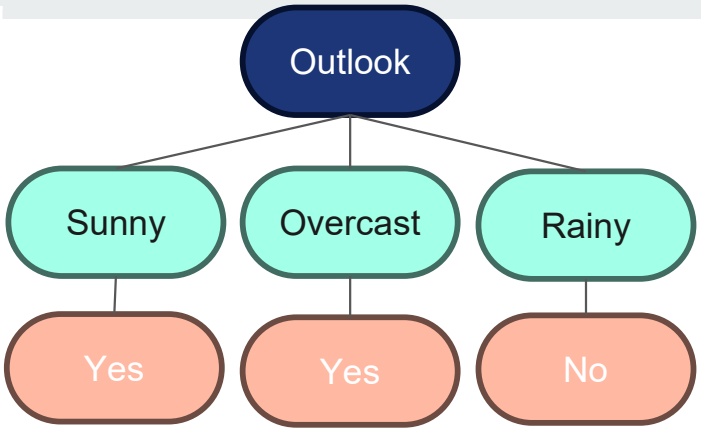
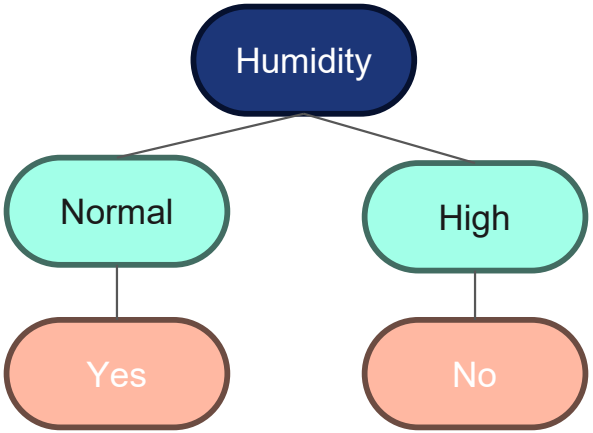
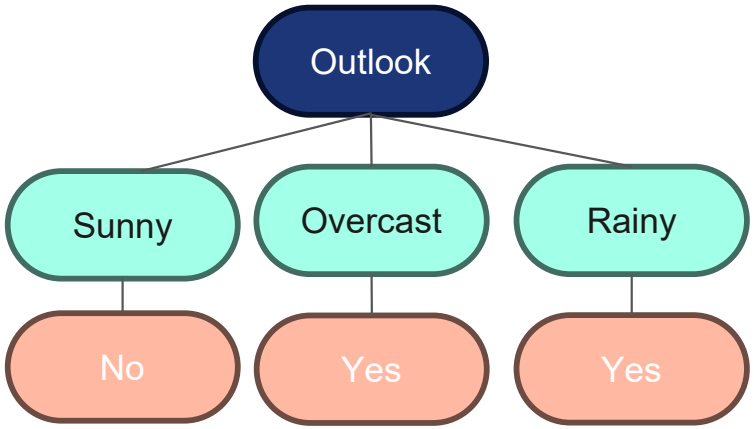


Tree 1



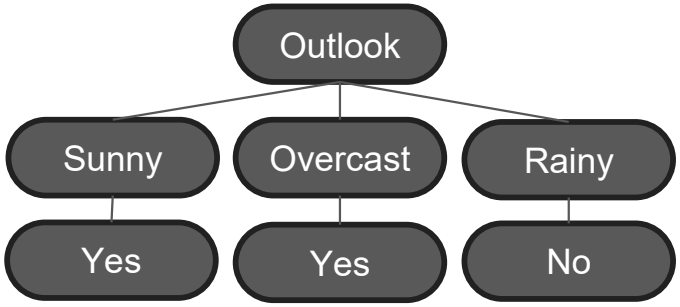
Tree 2



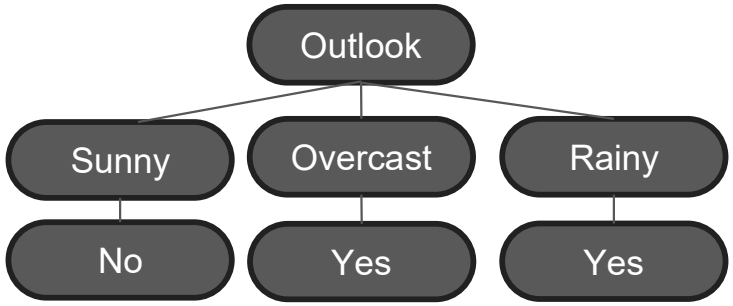
Tree 3

How does it take a decision?

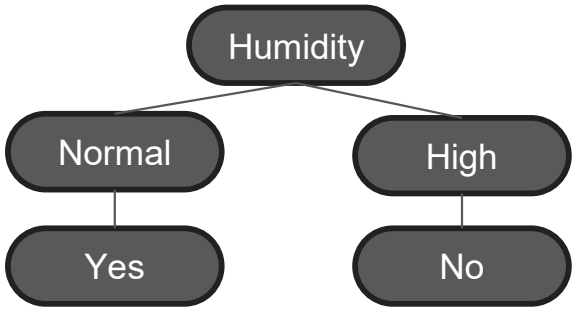
Outlook	Temp	Humidity	Windy	Play?
Rainy	Mild	High	False	???



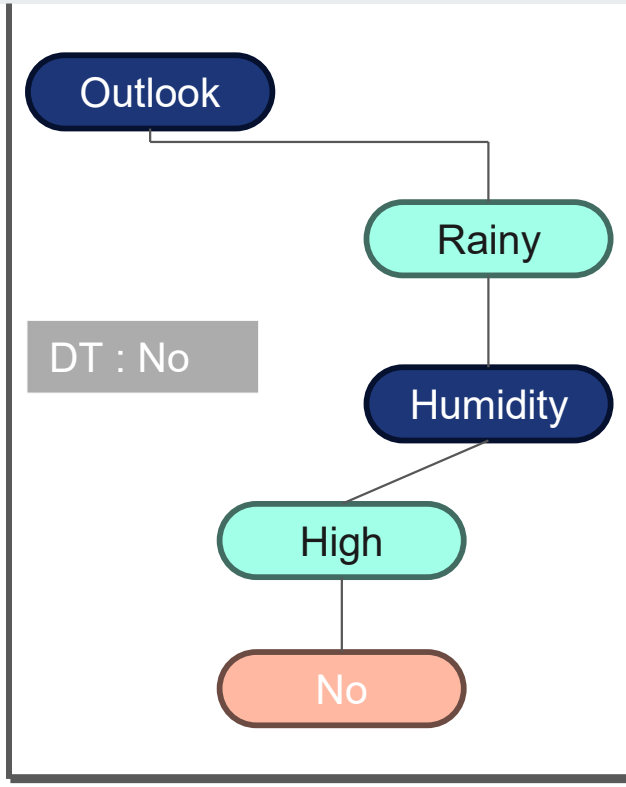
Tree 1



Tree 3



Tree 2

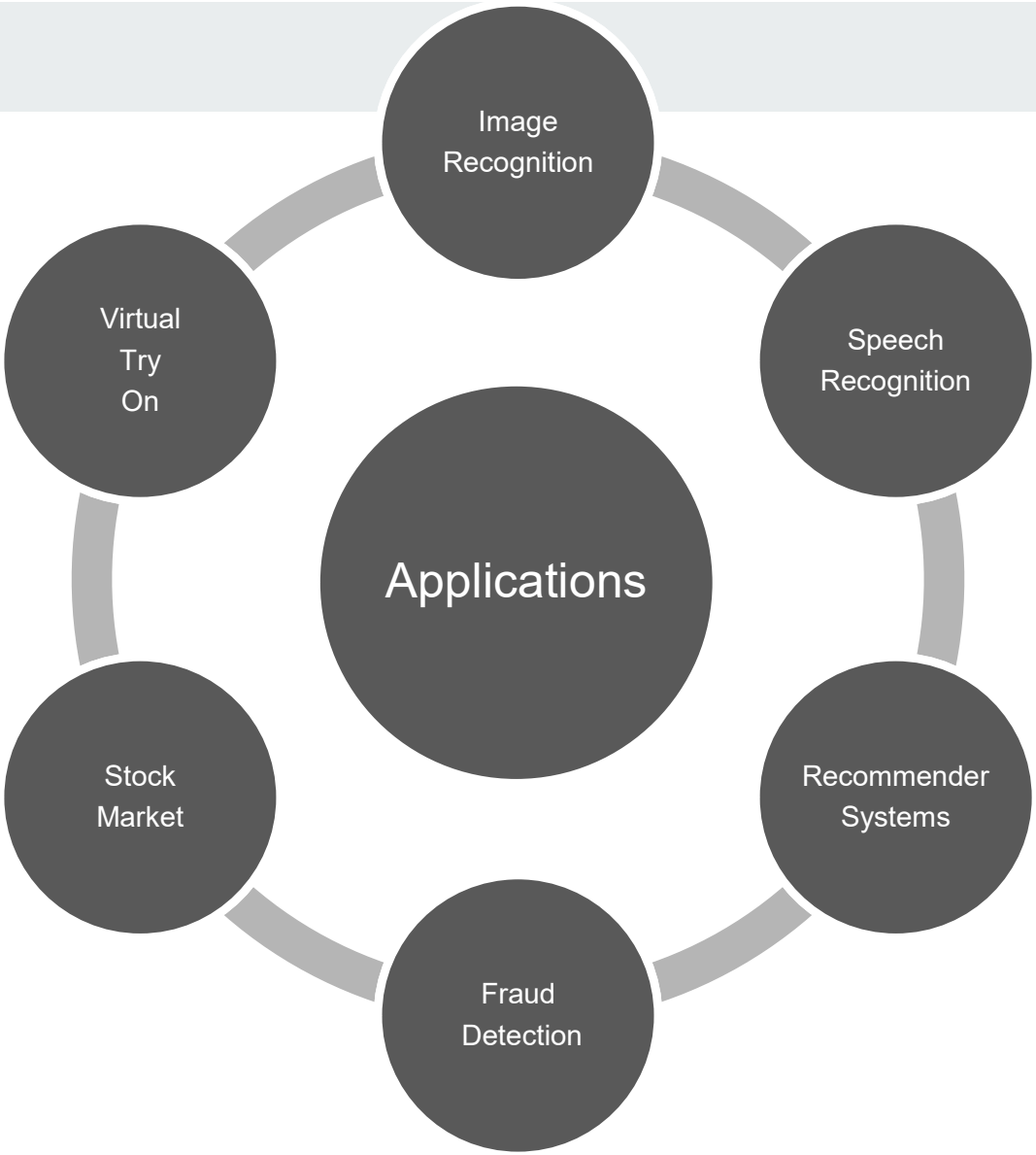


RF ➡

Tree 1 : No
Tree 2 : No
Tree 3 : Yes

Yes:1
No:2

Result : No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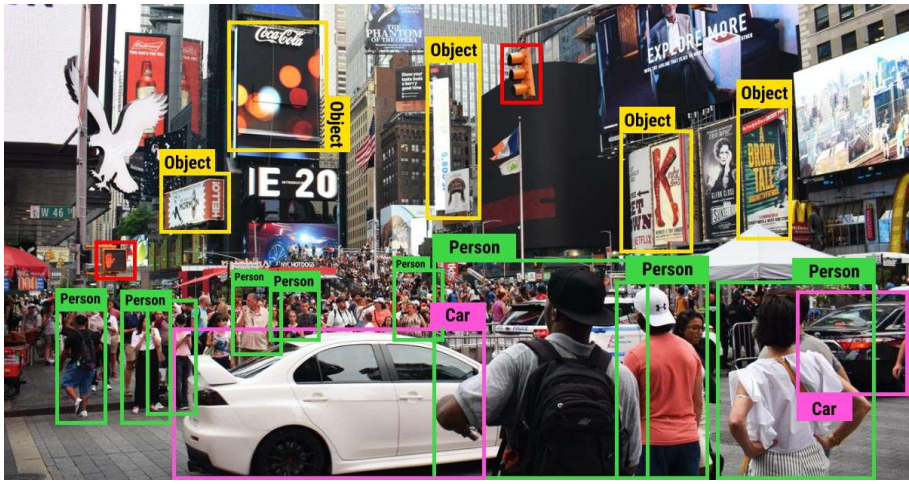
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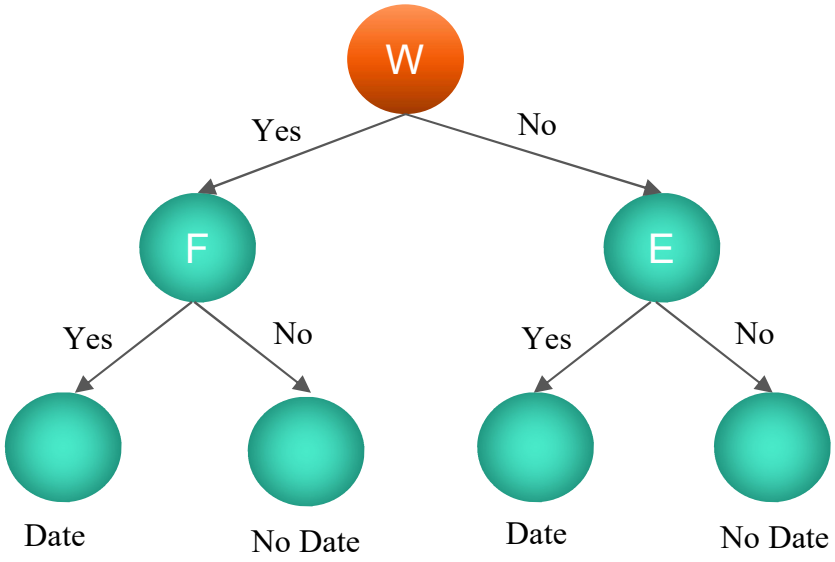

Weekend	Evening	Food	Date?
Yes	Yes	Yes	Yes
No	Yes	No	No
No	No	Yes	No
Yes	No	Yes	Yes
Yes	No	No	No
Yes	Yes	No	No